



RENEWABLE ENERGY TECHNOLOGY

SOLAR THERMAL PROJECT 2024

Solar Water Heating - Techno-Economic Assessment

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1. INTRODUCTION

Solar energy is one of the most abundant renewable resources which has been extensively used for power and hot water generation over the past decades. Solar water heating system (SWHS) is a dominant form of clean, sustainable and efficient solar energy application, which is able to capture the solar radiation and heat up water for domestic hot water use or space heating (Zhang, R. et al 2018). The performance of a solar water heater system is highly dependent on its orientation, optical and geometric properties, macro and microclimatic conditions, geographical position, operational parameters, and the period of use (Ertekin, C et al 2008). Solar water heating system (SWH) systems can operate independently or as a hybrid with electric or gas heating sources. Different types of SWH systems are used according to the locations, altitudes and weather conditions (Arif, M. et al 2022). The most commonly used solar water heating system for domestic needs is through natural circulation type that consists of a flat plate solar collector connected to an insulated storage tank. The sun's rays pass through the glass and are trapped in the space between the cover and plate or are absorbed by the black body. The circulating water through a conduit system located between the cover and absorber plate is heated and then carried to the storage tank. Flat plate collectors are most suitable when a temperature below 100 °C is required. These are simple to assemble; low cost; simple in design and fabrication; durable; do not require sun-tracking; can work on cloudy days; and require minimum maintenance (Ertekin, C et al 2008). This report presents a techno-economic assessment of implementing a solar water heating system for a single-family house located in Borrowdale, Harare City, Zimbabwe. The purpose of the analysis is to evaluate the feasibility of using solar energy to meet domestic hot water demand through energy output estimates, cost savings, and overall economic impact. The family home has four residents, and the rooftop area suitable for installing a solar collector is estimated at 10 m².

Types of solar water heater

There are two types of solar water heaters available: Active (based on flat plate collectors or FPC) and Passive (based on evacuated tube collectors or ETC). Active type solar water heaters use flat plate collectors, while Passive type solar water heaters use evacuated tube collectors. Flat plate collector (FPC) systems are metallic and have a longer lifespan, whereas evacuated tube collectors (ETC) are made of glass and are more fragile, giving them a shorter lifespan.

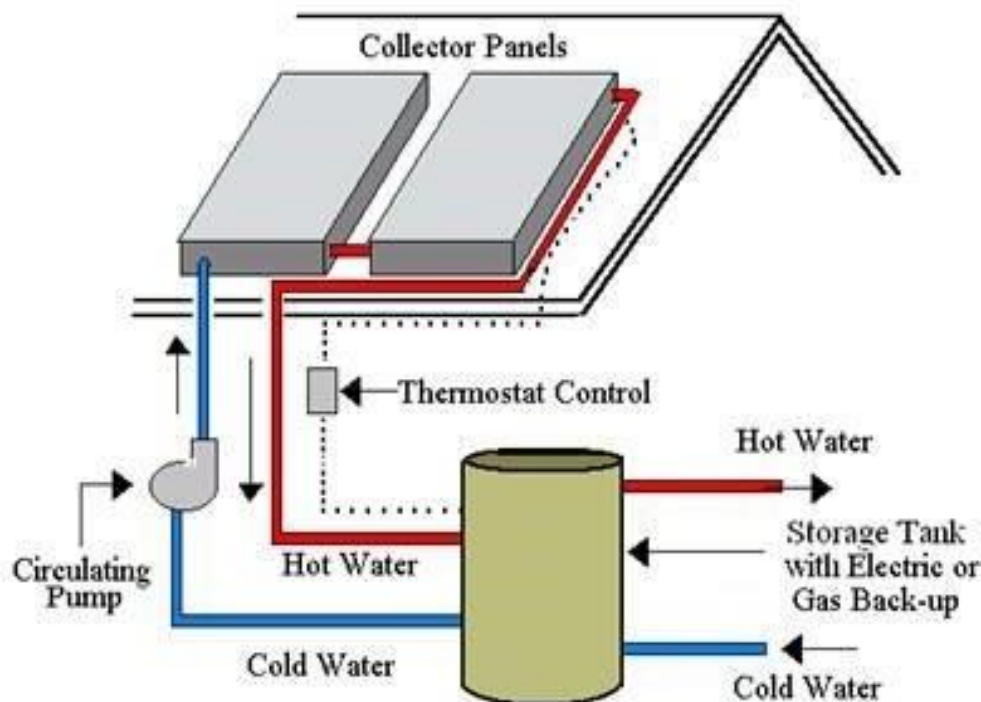


Figure 1: Solar Water Heating System (Priyadarshini, 2020)

Table 1 (Advantages & Disadvantages)

ADVANTAGES	DISADVANTAGES
Reduces energy bills	High initial costs
Environmentally friendly	Weather dependent
Energy independence	Requires space for installation
Utilizes renewable energy	Needs regular maintenance
Low maintenance costs	Limited efficiency in cold climates

2. OBJECTIVE

The primary objective of this study is to assess the technical and economic viability of installing a solar water heating system to meet the domestic hot water requirements of a single-family house in Borrowdale, Harare City, Zimbabwe. Specific objectives include:

1. Estimating the annual solar radiation—horizontal and tilted
2. Analyzing the total incremental initial costs, incremental O&M costs and fuel cost savings.
3. Estimating the system's energy output (energy saved) and solar fraction using location-specific parameters.
4. Determining the optimal system design parameters such as optimal tilt, facing, solar collector area, solar collector type, latitude, longitude and azimuth and storage capacity.
5. Evaluating key economic metrics, such as cost analysis—which includes initial costs, annual costs, and annual savings—emissions analysis—which includes the gross annual GHG emission reduction—financial analysis—which includes financial parameters, annual savings and revenue, NPV (Net Present Value), pre-tax IRR (Internal Rate of Return), pre-tax MIRR (Modified Internal Rate of Return), net yearly cash flow, simple payback, and equity payback—and sensitivity and risk analysis, which includes sensitivity range, thresholds, risk performance analysis, number of considerations, impact on equity payback, and distribution of equity payback.

3. METHODOLOGY

The methodology employed in this study includes the following steps:

RETScreen Software Analysis

RETScreen software was used to simulate the energy output, economic performance, and environmental impact of the proposed solar water heating system. The software allows for the modeling of clean energy technologies with built-in databases for climate and cost information.



Figure 2: RETScreen

Determining the Optimal Inclination Angle of the Solar Collectors

Using the 26 Optimal Tilt app as shows in Table 2 and Figure 3, the optimal tilt angle for solar collectors in Borrowdale, Harare City, Zimbabwe was determined to be 16.1°. This angle was selected to maximize solar energy capture throughout the year, considering the latitude of -17.756.

Facing Direction and Orientation of the Collector

Since Borrowdale is in the southern hemisphere, the solar collector is determined to face North for optimal solar energy collection. Additionally, the azimuth angle, which measures the deviation from true north, was set at 0° as shown in Table 2 and Figure 3, thus ensuring that the collectors are aligned for maximum solar exposure.

Latitude and Longitude

The exact geographic coordinates of the house in Borrowdale were used for RETScreen's climatic data inputs: Latitude: -17.756 and Longitude: 30.107.

Table 2

Location	Borrowdale, Harare City, Zimbabwe
Optimal tilt	16.1
Facing	North
Latitude	-17.756
Longitude	30.107
Azimuth	0

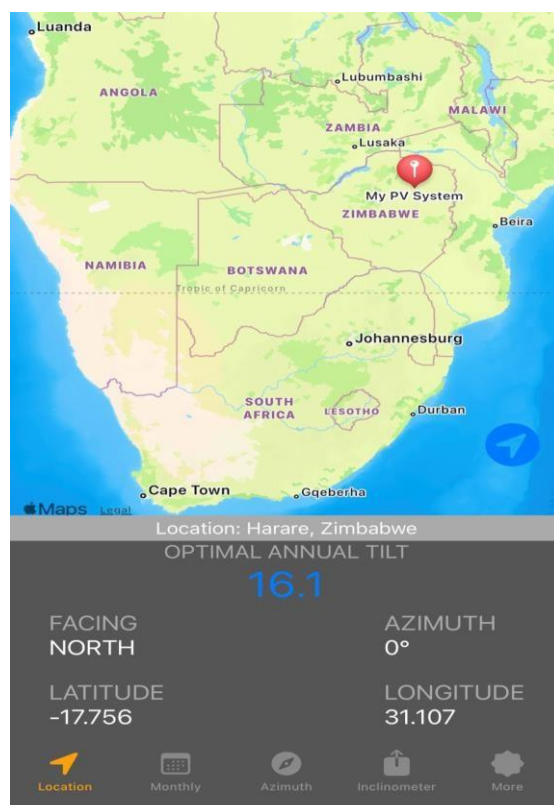
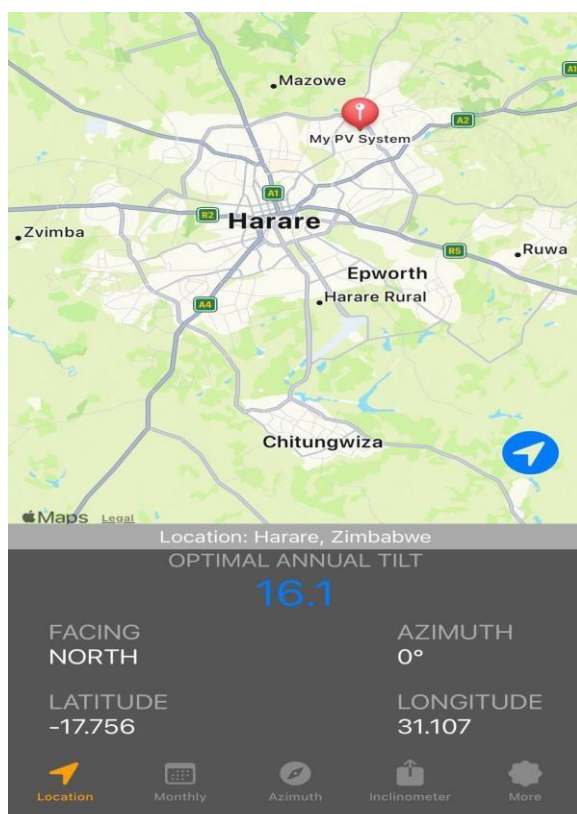


Figure 3: 26 Optimal Tilt app

Energy and Economic Calculations

RETScreen was also used to compute the expected energy output from the solar water heating system, based on local solar radiation data and the system's configuration. Economic calculations included estimates of installation costs, operational savings, and financial incentives. The payback period, NPV, and IRR were calculated based on these inputs, thus providing a comprehensive overview of the system's economic performance.

4. DATA SOURCE AND ASSUMPTIONS

1. Climatic and Solar Radiation Data: Data on solar radiation and local weather patterns for Borrowdale were sourced from the RETScreen software database, using the specified coordinates for the location
 - a. Latitude: -17.756
 - b. Longitude: 30.107
2. The seasonal efficiency was assumed to be 100%.
3. In the Load Type calculator, the number of units is set at 20, with an occupation rate of 90%.
4. RETScreen estimated daily hot water consumption to be 1080 L/d; however, the assumed value is determined to be 500 L/d. The temperature is expected to reach 70°C, and the house operates 24 hours a day. It is also assumed that no heat recovery efficiency is needed.
5. The solar tracking mode is assumed to be fixed. Since the desired temperature is relatively high, the assumption considers using a glazed solar collector to reduce heat losses. The F_r ($\tau \alpha$) coefficient is estimated to be 0.69, the F_r UL coefficient is 1.1, and the temperature coefficient for F_r UL is 0.01. The miscellaneous losses are estimated at 2%, with a storage capacity of 750 L. It is assumed that heat exchangers are not required, and additional miscellaneous losses are estimated at 1%. The pump power per solar collector area is assumed to be 0 W/m², and the system's initial cost is set at 300 ZWD.

5. MODEL CONSTRUCTION

The RETScreen model was constructed using the following data and assumptions:

- Location: Borrowdale, Harare City, Zimbabwe, with the specified geographic coordinates of Latitude: -17.756 and Longitude: 30.107.
- System Configuration: The model included a solar water heating system with a collector area of 10 m², installed at an optimal tilt angle of 16.1° and oriented to face North, with an azimuth of 0°.
- Energy Source Displacement: The system is expected to reduce electricity or gas consumption, depending on the family's current water heating source.
- Performance Factors: Solar radiation data, weather patterns, and system efficiency were incorporated into the model to estimate the annual energy output.

The RETScreen – Location

The RETScreen model was constructed using the inputs from Table 2 as shown below

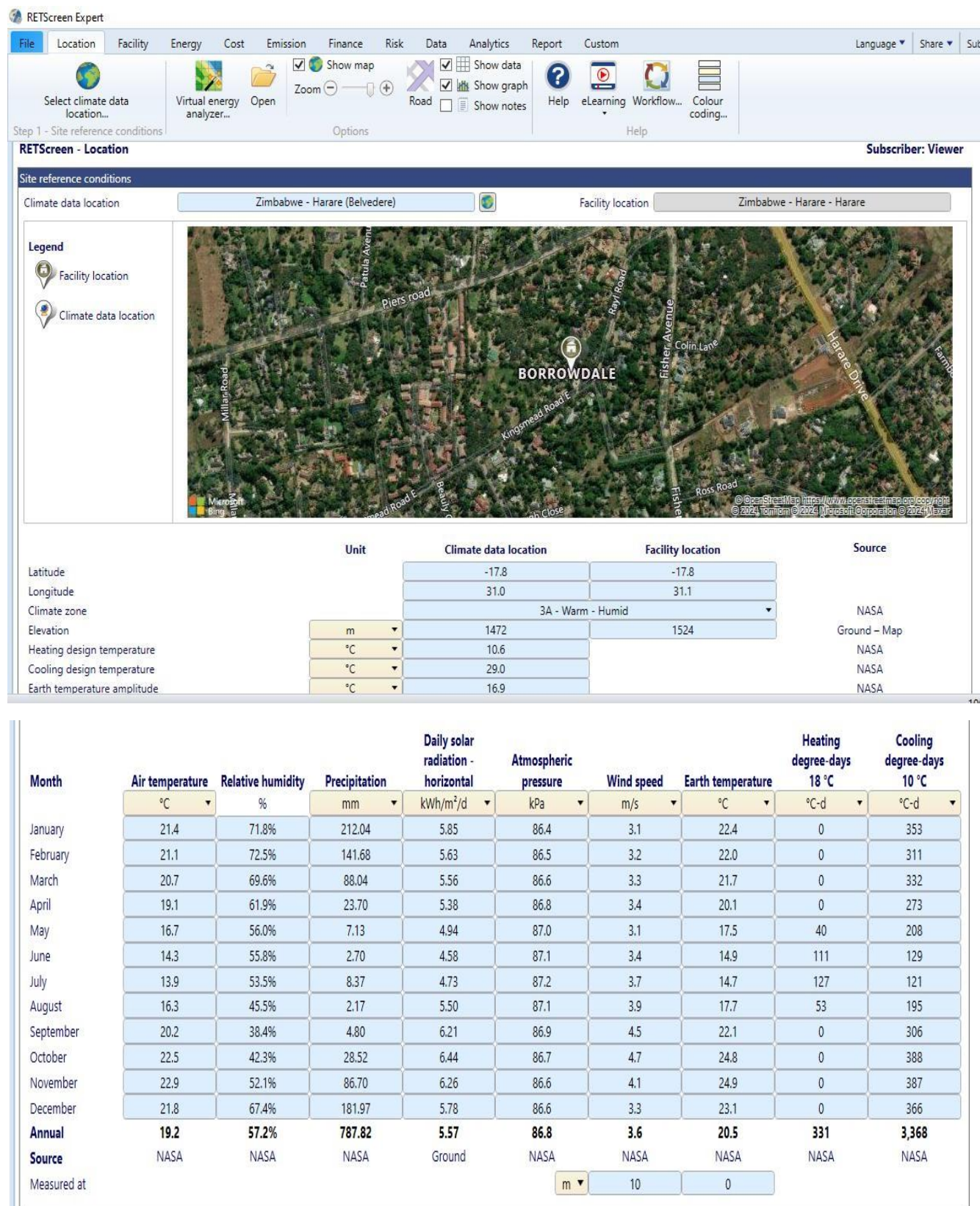


Figure 4: RETScreen – Location Data

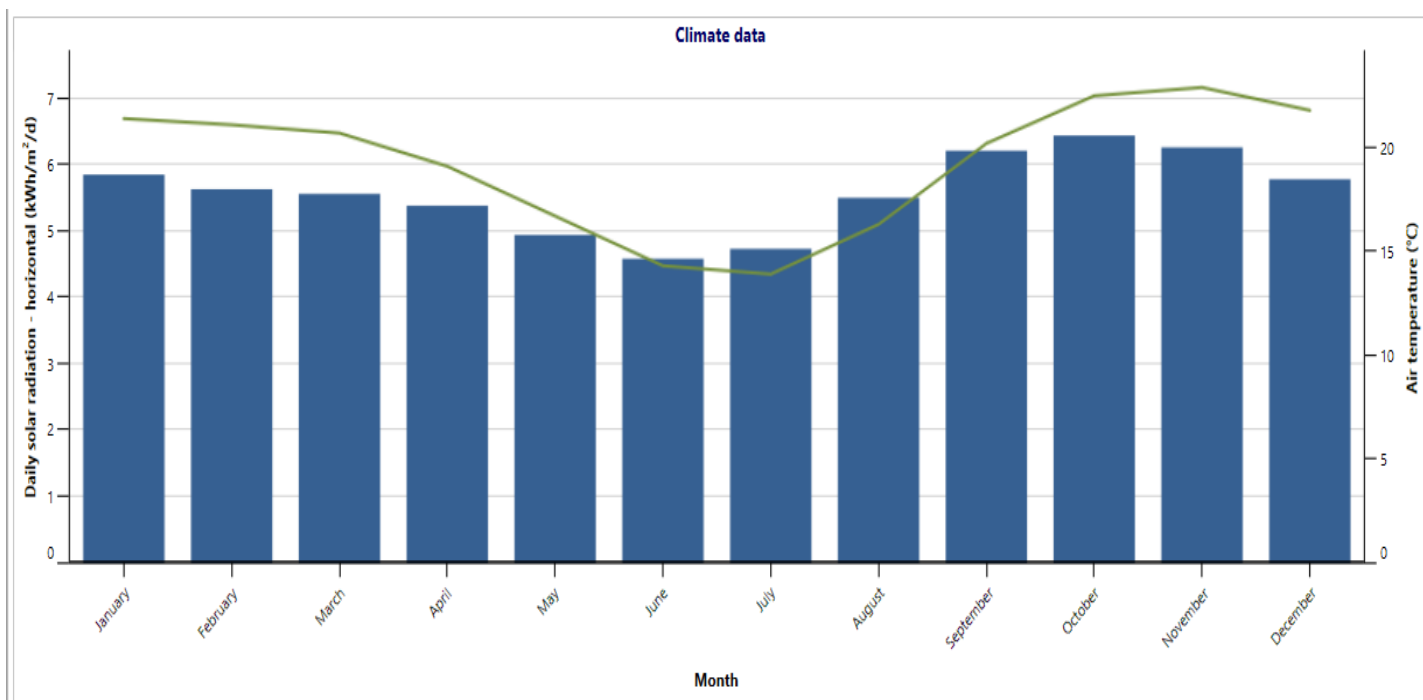


Figure 5: Climate Data

The Facility

The facility type was determined to be Individual Measure for a Solar Water Heater.

RETScreen - Facility
Subscriber: Viewer

Facility information

Facility type
Individual measure

Type
Solar water heater

Description
Laboratory

Prepared for
RETScreen Training Institute

Prepared by
CanmetENERGY - Varennes

Facility name
CanmetENERGY - Varennes

Address
1615 Boulevard Lionel-Boulet

City/Municipality
Harare

Province/State
Harare

Country
Zimbabwe

Figure 6: RETScreen Facility

The Energy Model

The heating system description was determined to be a Water Heater. The Base Case and Proposed Case were determined as the Electricity. The incremental initial cost and the incremental O & M savings were determined to be in ZWD (Zimbabwean Dollar).

Heating system

Description: Water heater

Note:

Method

Single fuel | Multiple fuels

Options

eLearning | RETScreen Connect

Heating system

Technology: Water heater

Fuel type: Electricity

Fuel rate: 0.094 ZWD/kWh

☐ Heating equipment

Seasonal efficiency: 100%

Incremental initial costs: ZWD

Incremental O&M savings: ZWD

Figure 7

For the cold-water supply, RETScreen formula method was used to estimate the temperature of the incoming cold water being fed into the house. The house operates 24 hours per day.

Hot water

Description: House Hot Water

Note:

Method

Method 1 | Method 2 | Swimming pool

Options

eLearning | RETScreen Connect

Hot water - Method 1

☒ Load type - calculator

Number of units: 20

Occupancy rate: 90%

Daily hot water use - estimated: 1,080 L/d

Hot water use: 500 L/d

Temperature: 70°C

Supply temperature method: Formula

Water temperature - minimum: 17.4°C

Water temperature - maximum: 20.5°C

Operating hours: 24 h/d

Heat recovery efficiency: %

☐ Percent of month used

Incremental initial costs: ZWD

Incremental initial costs - other: ZWD

Incremental initial costs - total: ZWD

Incremental O&M savings: ZWD

Heating system: Boiler - Electric

Heating: 10,810 kWh

Energy saved: 0

Figure 8

In the proposed case, the Solar Water Heater was identified as the heat supplier and used for optimization. For hot water, the House Hot Water system was selected. RETScreen then automatically confirmed the water temperature to be 70 degrees, with heating energy at 10,810 kWh. In the resource assessment, the solar setup was determined to be fixed, with an azimuth of 0 and a slope of 16.1 degrees. Since the desired temperature is high, a glazed solar collector was selected to reduce heat loss (Shukla et al 2013). The gross area and aperture area per solar collector were determined to be 10 m². RETScreen suggested the number of collectors be 1. For the balance of system option, storage was deemed necessary to ensure hot water supply during the night and on cloudy days.

RETScreen - Energy Model

Individual measure - Laboratory - Solar water heater

Fuels & schedules
 Electricity and fuels
 Schedules

Equipment
 Heating
 Boiler - Electric
 Domestic hot water - Electricity
 Domestic hot water - Natural gas
 Water heater
 Cooling
 Air conditioning

End-use
 Building envelope
 Office
 Laboratory

Solar water heater

Description: Solar water heater (1)

Note:

Solar water heater

Load characteristics

Hot water: House Hot Water

Temperature: °C 70

Heating: kWh 10,810

Resource assessment

Solar tracking mode: Fixed

Slope: 16.1

Azimuth: 0

Options

eLearning

RETScreen Connect

Show data

Month	Percent of month used - base case %	Percent of month used - proposed case %	Daily solar radiation - horizontal kWh/m ² /d	Daily solar radiation - tilted kWh/m ² /d	Heating delivered kWh
January	100%	100%	5.85	5.87	758.736
February	100%	100%	5.63	5.48	649.705
March	100%	100%	5.56	5.17	685.036
April	100%	100%	5.38	4.63	601.421
May	100%	100%	4.94	3.94	533.338
June	100%	100%	4.58	3.50	458.347
July	100%	100%	4.73	3.69	503.372
August	100%	100%	5.50	4.55	621.774
September	100%	100%	6.21	5.57	717.596
October	100%	100%	6.44	6.17	795.688
November	100%	100%	6.26	6.24	770.124
December	100%	100%	5.78	5.88	757.228
Annual	100%	100%	5.57	5.05	7,852.363
Annual solar radiation - horizontal	MWh/m ²	2.03			
Annual solar radiation - tilted	MWh/m ²	1.84			

Solar water heater

Figure 9



Type		Glazed ▼
Manufacturer		
Model		
Gross area per solar collector	m ² ▼	10
Aperture area per solar collector	m ²	10
Fr (tau alpha) coefficient		0.69
Fr UL coefficient	(W/m ²)/°C ▼	1.1
Temperature coefficient for Fr UL	(W/m ²)/°C ² ▼	0.01
Number of collectors - suggested		1
Number of collectors		1
Solar collector area	m ²	10
Capacity	kW	7
Miscellaneous losses	%	2%

Balance of system & miscellaneous

Storage	yes/no	Yes ▼
Storage capacity / solar collector area	L/m ² ▼	75
Storage capacity	L	750
Heat exchanger	yes/no	No ▼
Miscellaneous losses	%	1%
Pump power / solar collector area	W/m ² ▼	0
Electricity rate	ZWD/kWh	0.094
Initial costs	ZWD ▼	300
O&M costs (savings)	ZWD	

Summary

Electricity - pump	kWh ▼	0
Energy saved	kWh ▼	7,852
Solar fraction	%	72.6%

Figure 10

RETScreen full Energy Model

Show: All	Heating	Cooling	Electricity	Incremental initial costs	Fuel cost savings	Incremental O&M savings	Simple payback yr	Include measure?
Fuel saved	kWh	kWh	kWh	ZWD	ZWD	ZWD		☒
Heating								
Boiler - Electric	7,285			63,000	-240	3,142	21.7	☒
Domestic hot water - Electricity	0			0	0	0		☒
Domestic hot water - Natural gas	0			0	0	0		☒
Water heater	0			0	0	0		☒
Cooling								
Air conditioning		0		0	0	0		☒
Building envelope								
Office	3,077	9,365		3,100	1,005	0	3.1	☒
Laboratory	5,549			3,100	238	0	13.0	☒
Ventilation								
Office - M3	2,797	8,065		25,974	886	0	29.3	☒
Office - M30	832	2,399		7,727	264	0	29.3	☒
Laboratory - M2	1,221	3,521		11,339	387	0	29.3	☒
Laboratory - M1	1,400			13,000	66.1	0	196.8	☒
Laboratory - M21	1,135			10,538	53.6	0	196.8	☒
Laboratory - M20	1,092			10,143	51.5	0	196.8	☒
Laboratory - M11	1,101			10,221	51.9	0	196.8	☒
Laboratory - M4	1,540			14,300	72.7	0	196.8	☒
Laboratory - M5&6	1,036			9,620	48.9	0	196.8	☒
Laboratory - M8	196			1,820	9.2	0	196.8	☒
Laboratory - M31	1,540			14,300	72.7	0	196.8	☒
Office - M40	420	1,211		3,900	133	0	29.3	☒
Lights								
Office			34,532	11,400	3,229	180	3.3	☒
Exterior - Parking			4,319	3,615	404	-25	9.5	☒
Exterior - Facade			2,978	2,550	278	-130	17.2	☒
Exterior - Door			1,489	2,550	139	-130	276.0	☒
Laboratory			31,962	26,330	2,988	490	7.6	☒
Meeting room Lobby			3,915	7,230	366	79.6	16.2	☒
Office - 3			422	0	39.5	0	Immediate	☒

Figure 11

Electrical equipment									
Office	4,168		3,000	390	0	7.7		✓	
Server room	0		0	0	0			✓	
Laboratory	0		0	0	0			✓	
Cafeteria	4,505		400	421	50	0.8		✓	
Office - 3	757		500	70.8	0	7.1		✓	
Hot water									
Hot water - Electricity	0		0	0	0			✓	
Hot water - Natural gas	0		0	0	0			✓	
House Hot Water	0		0	0	0			✓	
House Hot Water	0		0	0	0			✓	
Pumps									
Space heating - Coils	0		0	0	0			✓	
Domestic hot water - Circulating pump	2,000		400	187	0	2.1		✓	
Boiler - Circulating pump	0		0	0	0			✓	
Space heating - Radiant system	0		0	0	0			✓	
Heater	0		0	0	0			✓	
Heating	0		0	0	0			✓	
Extension - Heating	0		0	0	0			✓	
Space cooling	0		0	0	0			✓	
Fans									
Office	1,675		0	157	0	Immediate		✓	
Washroom - Exhaust	35.1		0	3.3	0	Immediate		✓	
Office - M30	427		0	39.9	0	Immediate		✓	
Laboratory - M30	2,994		0	280	0	Immediate		✓	
Laboratory - M1	1,222		0	114	0	Immediate		✓	
Laboratory - M21	991		0	92.6	0	Immediate		✓	
Laboratory - M20	622		0	58.2	0	Immediate		✓	
Laboratory - M11	951		0	88.9	0	Immediate		✓	
Laboratory - M4	1,195		0	112	0	Immediate		✓	
Laboratory - M5&6	1,142		0	107	0	Immediate		✓	
Laboratory - M8	203		0	19	0	Immediate		✓	
Laboratory - M31	1,508		0	141	0	Immediate		✓	
Office - M40	188		0	17.6	0	Immediate		✓	
Showers - Exhaust	36.3		0	3.4	0	Immediate		✓	
Process electricity									
Standby losses - Laboratory	0		0	0	0			✓	
Forklift truck - Charging station	0		0	0	0			✓	
Vehicle - Charging station	0		0	0	0			✓	
Humidifier	0		0	0	0			✓	
Heating									
Solar water heater - Domestic hot water - Electricity	3,542	0	10,898	331	0	32.9		✓	
Solar water heater - Domestic hot water - Natural gas	1,146	0	2,724	57.2	0	47.7		✓	
Solar water heater	0	0	0	0	0			✓	
Solar water heater (1)	8,598	0	300	641	0	0.5		✓	
Power									
Photovoltaic - 40 kW	39,393		87,318	3,683	-1,548	40.9		✓	
Total	43,509	24,560	143,630	361,297	17,558	2,109	18.4		

Figure 12

6. RESULTS, ANALYSIS AND DISCUSSION

RESULTS (Table 3)

Location	Borrowdale, Harare City, Zimbabwe
Latitude	-17.756
Longitude	30.107
Azimuth	0
Optimal tilt	16.1
Facing	North
Solar tracking mode	Fixed
Solar type	Glazed
Gross area per solar collector	10 m ²
Annual solar radiation- horizontal	2.03 MWh/m ²
Annual solar radiation- tilted	1.84 MWh/m ²
Solar water heater capacity	7 kW
Base case & Proposed case Heating	10 810 kWh
Storage capacity/ solar collector area	75 L/m ²
Storage capacity	750 L
Energy saved	7 852 kWh
Solar fraction	72.6%
Total incremental initial costs	361 297 ZWD
Total incremental O&M costs	2 109 ZWD
Total fuel cost savings.	17 558 ZWD
Total annual costs	107 873 ZWD
Total annual savings and revenue	99 771 ZWD
Net yearly cash flow- Year 1	-8.101 ZWD
Simple payback	18.4 years
Gross annual GHG emission reduction	12.4%
GHG reduction cost	129 ZWD/tCO ₂
NPV (Net Present Value),	-121 626 ZWD
Pre-tax IRR (Internal Rate of Return)	
➤ Equity	-1.5%
➤ Assets	-6.5%
Pre-tax MIRR (Modified Internal Rate of Return)	
➤ Equity	0.51%
➤ Assets	-4.3%
Sensitivity and Risk Analysis (Sensitivity range)	25%
Sensitivity and Risk Analysis (Level of risk)	10%

The Cost Analysis

The cost analysis takes into consideration initial costs, annual costs, and annual savings. The initial costs consist of 260,357 ZWD for incremental initial costs. The annual costs amount to 82,628 ZWD, which includes an O&M cost saving of (3,657) ZWD and fuel costs of 86,285 ZWD for the proposed case. The annual savings from the base case fuel cost are 99,771 ZWD, indicating potential positive net savings after accounting for the initial and operating costs.

RETScreen - Cost Analysis

Initial costs (credits)	Unit	Quantity	Unit cost	Amount
Incremental initial costs			ZWD	260,357
▼ Show data				
- User-defined	cost ▼		ZWD	-
+				
Total initial costs			ZWD	260,357
Annual costs (credits)	Unit	Quantity	Unit cost	Amount
O&M costs (savings)	project		ZWD	(3,657)
▼ Show data				
Fuel cost - proposed case			ZWD	86,285
- User-defined	cost ▼		ZWD	-
+				
Total annual costs			ZWD	82,628
Annual savings	Unit	Quantity	Unit cost	Amount
Fuel cost - base case			ZWD	99,771
- User-defined	cost ▼		ZWD	-
+				
Total annual savings			ZWD	99,771

Figure 13: Cost Analysis

The Emission Analysis

The emission analysis from the RETScreen Expert tool compares the GHG emissions of a baseline electricity system in Zimbabwe. The base case emits 778 tonnes of CO₂ (tCO₂), while the proposed case reduces emissions to 681 tCO₂. This results in a gross annual GHG emission reduction of 96.1 tCO₂, equivalent to a 12.4% reduction. The chart illustrates this decrease, and it is equated to taking 17.6 cars or light trucks off the road, further emphasizing the environmental benefit of the proposed system.

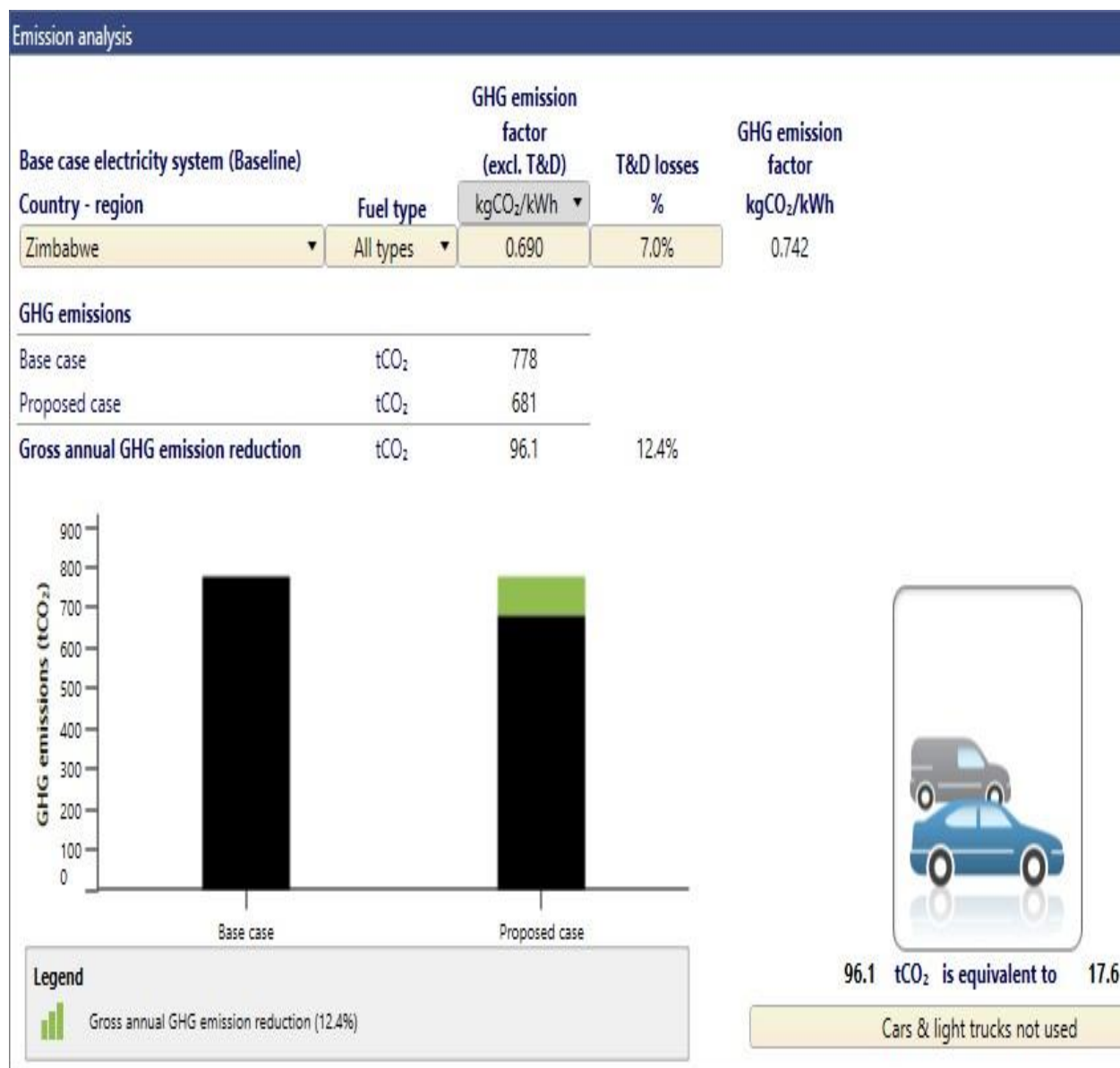


Figure 14: Emission Analysis

The Financial Analysis

The project has a negative net present value (NPV) of ZWD -121,626 and a low equity internal rate of return (IRR) of -1.5%, indicating poor financial viability. The benefit-cost ratio is 0.72, showing that the costs outweigh the benefits. With a simple payback period of 18.4 years, the project does not break even within its lifecycle. The initial cash flows are highly negative due to substantial upfront costs but begin to stabilize and become positive starting in year 13, reflecting the possibility of a delayed financial recovery.

RETScreen - Financial Analysis

Subscriber: Viewer

Financial parameters			Costs Savings Revenue			Yearly cash flows		
General			Initial costs			Year	Pre-tax	Cumulative
Fuel cost escalation rate	%	2%	Incremental initial costs	100%	ZWD	#	ZWD	ZWD
Inflation rate	%	2%				0	-108,389	-108,389
Discount rate	%	9%	Total initial costs	100%	ZWD	1	-7,708	-116,097
Reinvestment rate	%	9%				2	-7,307	-123,403
Project life	yr	20	Yearly cash flows - Year 1			3	-6,897	-130,301
Finance			Annual costs and debt payments			4	-6,480	-136,781
Incentives and grants	ZWD	0	O&M costs (savings)	ZWD	-2,109	5	-6,054	-142,835
Debt ratio	%	70%	Fuel cost - proposed case	ZWD	82,213	6	-5,620	-148,455
Debt	ZWD	252,908	Debt payments - 15 yrs	ZWD	27,768	7	-5,177	-153,632
Equity	ZWD	108,389	Total annual costs	ZWD	107,872	8	-4,725	-158,357
Debt interest rate	%	7%	Annual savings and revenue			9	-4,264	-162,621
Debt term	yr	15	Fuel cost - base case	ZWD	99,771	10	-3,794	-166,416
Debt payments	ZWD/yr	27,768	GHG reduction savings	ZWD	0	11	-3,315	-169,731
Income tax analysis			CE production revenue	ZWD	0	12	-2,826	-172,556
	☐		Total annual savings and revenue	ZWD	99,771	13	-2,327	-174,883
Annual savings and revenue			Net yearly cash flow - Year 1	ZWD	-8,101	14	-1,818	-176,701
GHG reduction savings			Financial viability			15	-1,299	-178,000
Gross GHG reduction	tCO ₂ /yr	128	Pre-tax IRR - equity	%	-1.5%	16	26,998	-151,002
Gross GHG reduction - 20 yrs	tCO ₂	2,564	Pre-tax MIRR - equity	%	0.51%	17	27,538	-123,464
GHG reduction savings	ZWD	0	Pre-tax IRR - assets	%	-6.5%	18	28,089	-95,375
Other revenue (cost)	☐		Pre-tax MIRR - assets	%	-4.3%	19	28,651	-66,724
Clean Energy (CE) production revenue	☐		Simple payback	yr	18.4	20	29,224	-37,500
			Equity payback	yr	> project			
			Net Present Value (NPV)	ZWD	-121,626			
			Annual life cycle savings	ZWD/yr	-13,324			
			Benefit-Cost (B-C) ratio		-0.12			
			Debt service coverage		0.72			
			GHG reduction cost	ZWD/tCO ₂	129			

Yearly cash flows

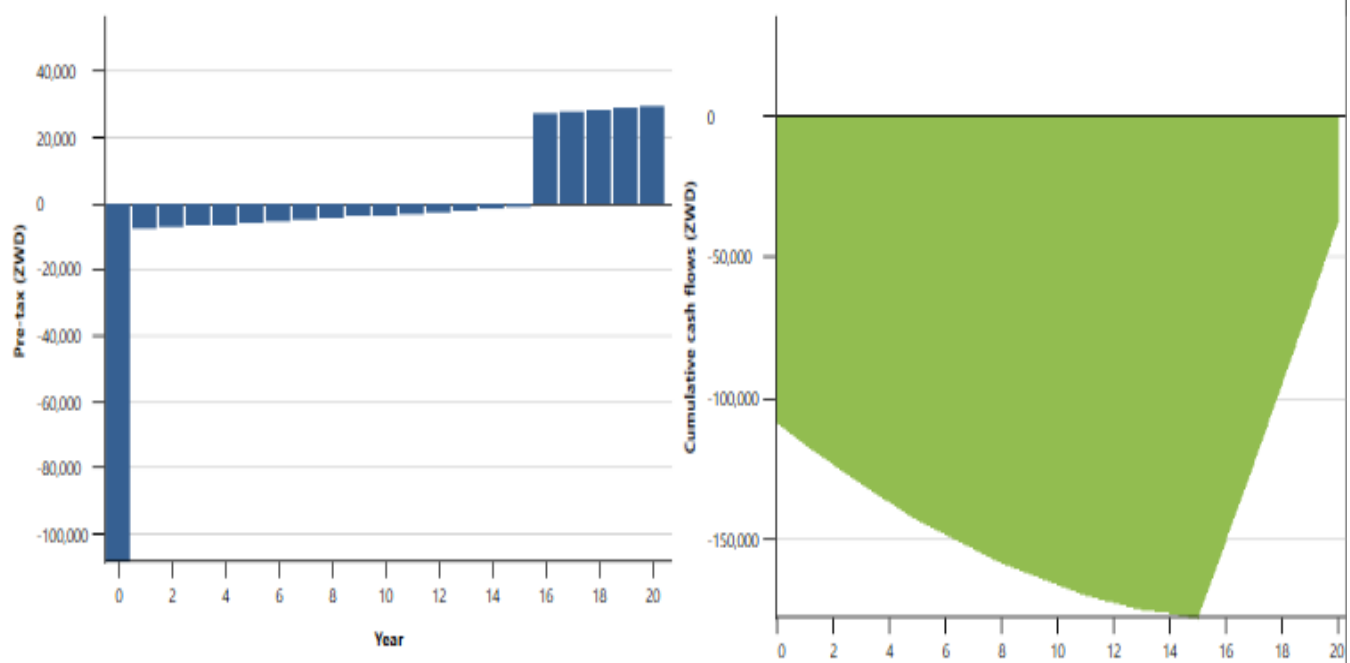


Figure 15: Financial Analysis

The Sensitivity and Risk Analysis

RETScreen - Sensitivity & Risk Analysis

Subscriber: Viewer

Sensitivity analysis

Perform analysis on

Equity payback

Sensitivity range

25%

Threshold

7

yr

- Remove analysis

Initial costs

ZWD

Fuel cost - base case

270,973

316,135

361,297

406,459

451,621

ZWD

-25.0%

-12.5%

0.0%

12.5%

25.0%

74,828

-25.0%

None

None

None

None

None

87,300

-12.5%

> project

> project

> project

> project

> project

99,771

0.0%

16.7

19.0

> project

> project

> project

112,243

12.5%

6.0

8.6

12.1

15.5

17.0

124,714

25.0%

3.2

4.2

5.5

7.1

9.2

- Remove analysis

Initial costs

ZWD

Fuel cost - proposed case

270,973

316,135

361,297

406,459

451,621

ZWD

-25.0%

-12.5%

0.0%

12.5%

25.0%

61,660

-25.0%

3.8

5.1

6.9

9.2

12.1

71,937

-12.5%

7.0

10.3

14.7

16.5

18.0

82,213

0.0%

16.7

19.0

> project

> project

> project

92,490

12.5%

> project

> project

> project

> project

> project

102,767

25.0%

None

None

None

None

None

- Remove analysis

Initial costs

ZWD

O&M

270,973

316,135

361,297

406,459

451,621

ZWD

-25.0%

-12.5%

0.0%

12.5%

25.0%

-1,582

-25.0%

17.1

19.5

> project

> project

> project

-1,845

-12.5%

16.9

19.3

> project

> project

> project

-2,109

0.0%

16.7

19.0

> project

> project

> project

-2,373

12.5%

16.5

18.8

> project

> project

> project

-2,636

25.0%

16.3

18.6

> project

> project

> project

- Remove analysis

Initial costs

ZWD

O&M

270,973

316,135

361,297

406,459

451,621

ZWD

-25.0%

-12.5%

0.0%

12.5%

25.0%

-1,582

-25.0%

17.1

19.5

> project

> project

> project

-1,845

-12.5%

16.9

19.3

> project

> project

> project

-2,109

0.0%

16.7

19.0

> project

> project

> project

-2,373

12.5%

16.5

18.8

> project

> project

> project

-2,636

25.0%

16.3

18.6

> project

> project

> project

- Remove analysis

Debt interest rate

%

Debt ratio

5.25%

6.13%

7.00%

7.88%

8.75%

%

-25.0%

-12.5%

0.0%

12.5%

25.0%

53%

-25.0%

18.7

19.3

19.9

> project

> project

61%

-12.5%

19.2

19.9

> project

> project

> project

70%

0.0%

19.7

> project

> project

> project

> project

79%

12.5%

> project

> project

> project

> project

> project

88%

25.0%

> project

> project

> project

> project

> project

- Remove analysis

Debt interest rate

%

Debt term

5.25%

6.13%

7.00%

7.88%

8.75%

yr

-25.0%

-12.5%

0.0%

12.5%

25.0%

11

-25.0%

18.6

19.2

19.7

> project

> project

13

-12.5%

19.2

19.8

> project

> project

> project

15

0.0%

19.7

> project

> project

> project

> project

17

12.5%

> project

> project

> project

> project

> project

19

25.0%

> project

> project

> project

> project

> project

Figure 16: Sensitivity Analysis

The risk analysis for the equity payback period was conducted using 500 different combinations of input parameters. The parameters varied include initial costs, operation and maintenance (O&M), fuel costs (proposed and base case), debt interest rate, and debt term. Each parameter was adjusted within a range of $\pm 25\%$ to observe its impact on the equity payback. The results show a median payback period of 16 years with a 10% risk level. The minimum payback period, with 90% confidence, is 7.2 years, while the maximum is 19.5 years. Overall, the analysis demonstrates a moderate risk in achieving equity payback within the estimated timeframe.

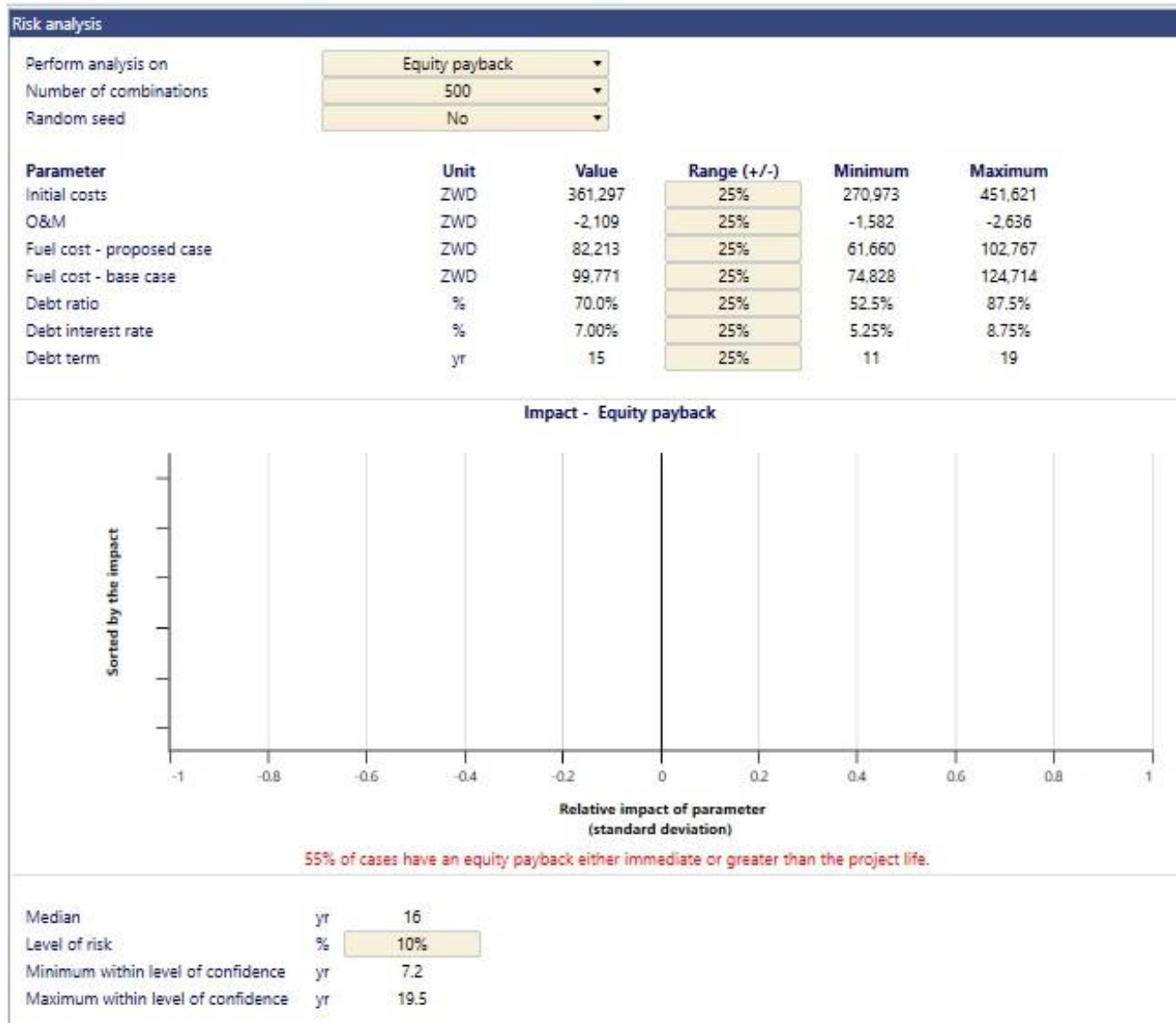


Figure 17: Risk Analysis

The impact graph reveals how the different parameters influence the equity payback period. The chart shows that none of the parameters have an extreme impact on the equity payback, as all the variations appear close to the baseline (centered around 0). This suggests that no single factor overwhelmingly determines the payback period. Instead, all parameters contribute relatively equally to the variability in payback times, and the financial outcome is sensitive to changes across all variables.

The distribution graph shows the spread of equity payback periods across the 500 simulations. The majority of payback periods fall between 14 and 18 years, with a peak around 17.8 years where approximately 6-7% of cases are concentrated. Notably, 55% of the simulations show a payback period either at or greater than the project life, indicating a significant likelihood that the equity may not be fully paid back within the desired timeframe. This poses a potential risk to project stakeholders, as over half of the outcomes suggest a delayed or extended payback period.

Distribution - Equity payback

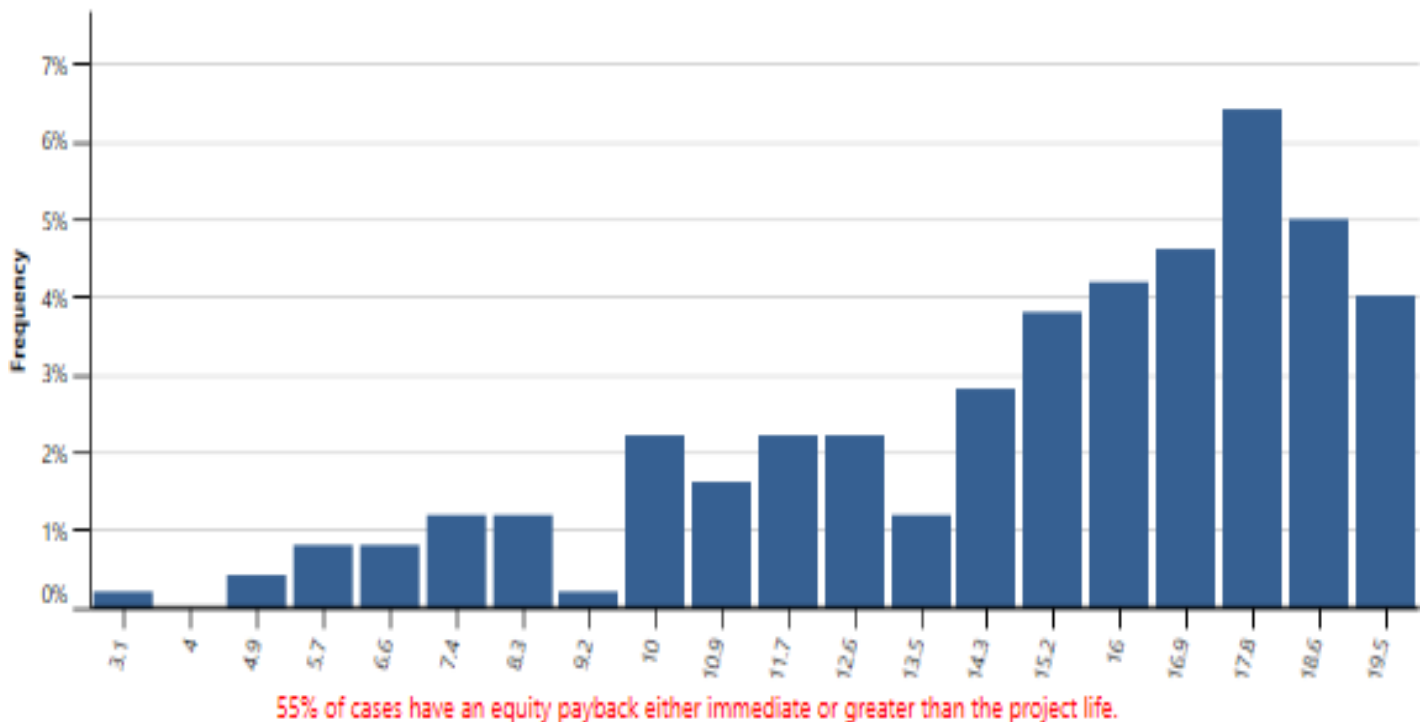


Figure 18: Distribution- Equity Payback

7. CONCLUSIONS

The techno-economic assessment of the solar water heating system for the single-family house in Borrowdale, Harare City, demonstrates both technical feasibility and economic benefits.

- ✓ **Energy Output:** The system is designed to generate 7,852 kWh annually, covering approximately 72.6% of the household's hot water demand, making it a significant contributor to energy efficiency.
- ✓ **Cost Savings:** The system offers notable cost savings, reducing fuel expenses by ZWD 107,873 and generating total annual savings and revenue of ZWD 99,711. This leads to a net annual cash flow of ZWD -8,101 in Year 1, indicating that initial returns are modest but improve over time.
- ✓ **Payback Period:** The simple payback period is estimated at 18.4 years. Although this period is relatively long, it reflects the system's potential for delayed financial recovery, as seen by the stabilization of cash flows after year 13.
- ✓ **Net Present Value (NPV):** The project has a negative NPV of ZWD -121,626, suggesting that while the system is technically sound, it may not provide immediate financial returns based on current costs and market conditions.
- ✓ **Internal Rate of Return (IRR):** The pre-tax IRR for equity is -1.5%, and for assets, it is -6.5%. Despite these low values, the modified internal rate of return (MIRR) for equity stands at 4.3%, showing some improvement in return when accounting for reinvestment rates.
- ✓ **Environmental Impact:** The system offers an annual greenhouse gas emission reduction of 129 ZWD/tCO₂, contributing positively to the environmental goals of the household.

Overall, while the project presents a long payback period and a negative NPV, the analysis demonstrates that solar water heating offers long-term environmental benefits and potential cost savings. With moderate financial risk and room for improvement through policy incentives or reduced upfront costs, this solar water heating system can serve as a sustainable option for households aiming to reduce reliance on traditional energy sources.

8. REFERENCES

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